Arnav Kirti

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EDUCATION

Indian Institute of Technology, Roorkee | *B.Tech, Metallurgical & Materials Engineering* *CGPA: 7.512* | *Aug 2023 - Present*
Class XII, Our Lady of Fatima Sr. Sec. School *90.60%* | *2022*
Class X, Our Lady of Fatima Sr. Sec. School *92.33%* | *2020*

EXPERIENCE

Open Source Contributor | *Summer of Bitcoin @Shopstr* *Remote* | *May 2026 - Aug 2026*
- Ranked **#2 all-time contributor** to Shopstr with 93 commits and 39K+ lines of code added to the codebase.
- Built a **GitHub Actions CI merge gate** for Shopstr, a Bitcoin/Nostr marketplace, blocking PRs that fail Jest tests.
- Scaled the test suite to **2,610 passing Jest tests** across Nostr order messaging, Cashu wallet, cache & NIP-99 parsers.
- Raised statement coverage from a low baseline to **98%** on parsers & ∼90% on Nostr signers; most helpers at 100%.
- **Tools Used:** TypeScript, Next.js, Jest, GitHub Actions, Nostr, Cashu.

Blockchain Developer Intern | *BlockseBlock* *Remote* | *June 2025 - July 2025*
- Developed and deployed decentralized applications (DApps) on the Internet Computer (ICP) blockchain.
- Designed and implemented smart contracts with a focus on scalability, security, and seamless user experience.
- Collaborated on integrating blockchain logic with interactive front-end interfaces for real-time on-chain interactions.
- **Tools Used:** Rust, React, Motoko, ICP SDK, JavaScript, Web3 APIs.

Full Stack Developer Intern | *M2ai* *May 2025 - July 2025*
- Integrated responsive front-end components with RESTful APIs, reducing user interaction latency by up to 50%.
- Collaborated on building scalable features using React.js and AWS, contributing to production-ready modules.
- Integrated **Stripe** payment gateway, enabling secure transactions and reducing payment processing time by 40%.
- **Tools Used:** React.js, AWS, Stripe, REST APIs.

PROJECTS

Enterprise Middleware & API Integration Platform | **Tech GC'25 (ICICI PS)** Link | *Apr 2025*
- Built a universal API framework for ERP integration across SAP, Oracle, & MS Dynamics and reducing integration time by 60%.
- Automated data mapping and dynamic update systems, allowing seamless transformation of ERP data into a unified schema.
- **Tools Used:** Node.js, Python, Express.js, React, OAuth2.0, JSON Schema, Shell, Machine Learning.

Data Compression & De-Compression Portal | **Models and Robotics Section** Link | *June 2025*
- Built a file compression platform supporting Huffman, RLE, and LZ77 algorithms, reducing file sizes by up to 40%.
- Designed an intuitive UI with algorithm education and data visualizations, improving user engagement.
- **Tools Used:** React 19, TypeScript, NestJS, Node.js, Vite, Tailwind CSS, Multer, Chart.js, Axios

Decentralised Social Media | **Blockchain Society IIT Roorkee** Link | *Jan 2024 - Feb 2024*
- Developed a decentralized social media platform where users can share tweets and interact with others.
- Integrated a Twitter timeline scraper into the frontend to display live feeds.
- **Tools Used:** Solidity, React.js, Node.js, Ether.js, Metamask, Material UI, CSS

Crypto Advisor | **Mira Hackathon** Link | *Jan 2025*
- Built a cryptocurrency investment platform offering personalized advice & diversification strategies with AI analysis.
- Integrated **Mira Flows** for AI-driven investment insights and **CoinGecko API** for real-time market data.
- **Tools Used:** React.js, Mira Flows, CoinGecko API.

ACHIEVEMENTS

- **ETHGlobal Agentic Ethereum Hackathon** - Coinbase AgentKit Pool Prize Winner for Defi-Dojo.
- Winner of **Coinbase Smart Contract Bounty**, among the Top 50 people to receive a $100 Amazon Gift Card.
- **StackTooDeep Hackathon** Winner for Decentralized Social Media Project.
- **StackTooDeepV2.0 Hackathon** Winner for Aura-Land Project.
- **Open Source Contributions:** Contributed to projects like GetAlby, Shopstr. Notable contributions: here.
- Secured **2nd Position** in **General Championship Tech** for High Prep (AI/ML) PS and Blockchain PS.
- Participated in the **MIRA Hackathon** (SDSLabs) and **ETHGlobal New Delhi**.

POSITIONS OF RESPONSIBILITY

Joint Secretary, Development Vertical | *Cultural Council, IIT Roorkee* *Nov 2025 - Aug 2026*
- Led the Development Vertical of Cultural Council IIT Roorkee, owning web projects under the GenSec Cultural Affairs.
- Developed the official Cultural Council IIT Roorkee website as the Development Vertical's flagship project.
- Ran technical interviews to recruit developers and mentored the new team in web development fundamentals.

Frontend Developer | *Appetizer, IIT Roorkee* *Aug 2024 - June 2025*
- Revamped the user interface for rebate and coupon workflows, reducing user friction and increasing daily usage by 30%.
- Integrated real-time menu & coupon features across hostels, enhancing app engagement and boosting rebates by 40%.
- Optimized React component structure and API calls, improving load speed by ∼45% and improving responsiveness.

Undergraduate Teaching Assistant | *Academic Reinforcement Program, IIT Roorkee* *Jan 2025 - May 2025*
- Mentored BSMS Physics juniors in CSE-101 (DSA), delivering study strategies and custom material.
- Selected based on my A+ performance in CSE-101, ensuring high technical and academic credibility in peer learning.

SPORTS

- **Taekwondo:** Black Belt (2nd Dan) | **National Championship:** Gold (2017), 1st Runner Up (2019), 2nd Runner Up (2024).
- **Sangram (Taekwondo)** - Gold Medalist | **BITS Open Sports Meet (Taekwondo)** - Winner | **Tug of War, Inter Bhawan Championship** - 2X Winner.